

Data Cleaning Steps Using Power Query

Cafe Sales Data – Documentation of Cleaning & Processing Steps

Overview

This document outlines the M-code steps used in Power Query / Power BI to clean and process the Cafe Sales Data dataset. Each section below corresponds to a stage in the cleaning pipeline, along with the relevant M-code snippet.

1. Loading the Data and Replacing Error Values

The data is first loaded from a CSV source, and the headers are promoted from the first row of the dataset.

```
let
    Source = Csv.Document(File.Contents("...")),
    #"Promoted Headers" = Table.PromoteHeaders(Source,
        [PromoteAllScalars=true])
```

Next, placeholder/error values such as N/A, UNKNOWN, and ERROR are identified and replaced with null, since these do not represent valid data and need to be handled consistently before further processing.

```
#"Replaced Value" = Table.ReplaceValue(..., "ERROR", null,
    Replacer.ReplaceValue, {...}),
#"Replaced Value1" = Table.ReplaceValue(..., "UNKNOWN", null, ...),
#"Replaced Value2" = Table.ReplaceValue(..., "N/A", null, ...),
...
```

2. Correcting Data Types

Each column is converted to its appropriate data type so that calculations and filtering behave correctly:

- Quantity → Whole Number (Int64.Type)
- Price Per Unit → Decimal Number (type number)
- Transaction Date → Date (type date)

```
#"Changed Type1" = Table.TransformColumnTypes(#"Replaced Value4",{
    {"Quantity", Int64.Type},
    {"Price Per Unit", type number},
    {"Transaction Date", type date}
})
```

3. Handling Null Values (Imputation)

Null values in key columns are filled with sensible default/representative values so the dataset remains complete for analysis:

- Item → "Juice"
- Payment Method → "Digital Wallet"
- Location → "Takeaway"
- Quantity → 3 (a representative/default value)

```
#"Replaced Value5" = Table.ReplaceValue(..., null, "Juice", ...,
{"Item"}),
#"Replaced Value8" = Table.ReplaceValue(..., null, 3, ...,
{"Quantity"})
```

4. Recalculating the "Total Spent" Column

To ensure accuracy, the original Total Spent column is removed and recalculated from the cleaned Price Per Unit and Quantity columns, then renamed back to its original name.

```
#"Removed Columns" = Table.RemoveColumns(...,{"Total Spent"}),
#"Inserted Multiplication" = Table.AddColumn(..., "Multiplication",
each [Price Per Unit] * [Quantity], type number),
#"Renamed Columns" = Table.RenameColumns(...,{"Multiplication",
"Total Spent"})
```

5. Removing Duplicates

Finally, any duplicate rows are removed from the table to ensure each transaction record is unique, and the resulting clean table is returned as the output of the query.

```
#"Removed Duplicates" = Table.Distinct("#Renamed Columns")
in
#"Removed Duplicates"
```